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SUPERSEDING
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MILITARY SPECIFICATION

HEATER, IMMERSION, M67, LIQUID FUEL FIRED, FOR Corrugated CANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a liquid fuel burning immersion-type heater designed primarily to heat water in a 32-gallon corrugated can.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards! and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

P-D-680 - Dry Cleaning Solvent
GG-P-455 - Plates and Foils, Photographic (Photosensitive Anodized Aluminum)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A Fsc 4540

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QQ-B-626 - Brass, Leaded and Nonleaded: Rod, Shapes, Forgings, and Flat Products with Finished Edges (Bar and Strip)
QQ-B-654 - Brazing Alloys, Silver
QQ-C-390 - Copper Alloy Castings (Including Cast Bar)
QQ-L-201 - Lead Sheet
QQ-s-635 - Steel Plate, Carbon
QQ-S-698 - Steel Sheet and Strip, Low-Carbon
QQ-S-781 - Strapping, Steel, and Seals
QQ-W-321 - Wire, Copper Alloy
RR-C-82 - Cans, Ash and Garbage, and Covers
RR-C-271 - Chains and Attachments, Welded and Weldless
TT-C-490 - Cleaning Methods for Surfaces and Pretreatment for Organic Coatings
TT-E-529 - Enamel, Alkyd, Semigloss
TT-P-636 - Primer Coating, Alkyd, Wood and Ferrous Metal
WW-P-404 - Pipe, Steel (Seamless and Welded, Black and Zinc Coated) (Galvanized)
W-P-521 - pipe Fittings, Flange Fittings, and Flanges: Steel and Malleable Iron (Threaded and Butt-Welded), Class 150
PPP-B-585 - Boxes, Wood, Wirebound
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-T-60 - Tape: Packaging, Waterproof
PPP-T-97 - Tape, Pressure-Sensitive Adhesive, Filament Reinforced

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MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-P-551 - Pipes and Elbows, Air Conditioning - Heating
MIL-R-3065 - Rubber, Fabricated Products
MIL-L-3150 - Lubricating Oil, Preservative, Medium
MIL-C-10382 - Corrosion Preventive, Petrolatum, Spraying Application: for Food Handling Machinery and Equipment
MIL-L-21260 - Lubricating Oil, Internal-Combustion Engine, Preservative and Break-In
MIL-C-52950 - Crate, Wood, Open and Covered

STANDARDS

FEDERAL

FED-STD-101 - Test Procedures for Packaging Materials
FED-STD-595 - Colors

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-417 - Classification System and Tests for Solid Elastomeric Materials (Symbols and Tests)

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a Part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-13-1874 - Heater, Immersion, M67, Liquid Fuel Fired, Cans, Corrugated; Containers for Level "A" Shipment
5-13-2679 - Assembly Complete
5-13-2680 - Heater Body Assembly
5-13-2681 - Heater Body Details
5-13-2682 - Miscellaneous Parts
5-13-2683 - Burner Assembly
5-13-2684 - Igniter Assembly and Name Plate
5-13-2685 - Fuel Tank Assembly
5-13-2686 - Valve and Filler Plug Assembly
5-13-2687 - Hanger Details

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation, Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

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AMERICAN WELDING SOCIETY (AWS)

Standard Qualification Procedure of the American Welding Society

(Copies should be obtained from the American Welding Society, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Welding Qualification of the American Society of Mechanical Engineers

(Copies should be obtained from the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 48 - Coatings, Gray, Iron
A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip
A 313 - Chromium - Nickel Stainless and Heat-Resisting Steel Spring Wire
A 511 - Seamless Stainless Steel Mechanical Tubing
A 513 - Electric-Resistance Welded Carbon and Alloy Steel Tubing
A 519 - Seamless Carbon and Alloy Steel Mechanical Tubing
A 575 - Merchant Quality Hot-Rolled Carbon Steel Bars
A 684 - Steel, Carbon, Strip, Cold Rolled Soft, Untempered Spring Quality
B 633 - Electrodeposited Coatings of Zinc on Iron and Steel
D 3951 - Standard Practice for Commercial Packaging

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

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* 3.2 Materials and components. Materials and components shall be as specified herein and as shown on the applicable drawings. Materials and components not definitely specified shall be of the quality normally used by the-manufacturer provided the completed items comply with all the provisions of this document (see 6.4).

3.2.1 Steel, carbon.

3.2.1.1 Steel, carbon sheet and strip. Carbon steel sheet and strip shall conform to QQ-S-698.

3.2.1.2 Bar stock. Cold finished carbon steel stock shall conform to QQ-S-635. Hot rolled steel bar stock shall conform to ASTM A 575.

3.2.1.3 Seamless steel tubing. Hot finished or cold worked finished seamless steel tubing shall conform to ASTM A 519.

3.2.1.4 Welded steel tubing. As-welded hot or cold rolled steel tubing shall conform to ASTM A 513.

3.2.1.5 Spring steel. Spring carbon steel strip shall conform to soft type, annealed, of ASTM A 684.

3.2.2 Corrosion-resisting steel.

3.2.2.1 Stainless steel sheet and strip. Stainless steel sheet and strip shall conform to type 304 and type 305 of ASTM A 167.

* 3.2.2.2 Corrosion-resisting steel wire. Corrosion-resisting steel wire shall conform to type 304 of ASTM A 313.

3.2.3 Stainless steel tubing. Stainless steel tubing shall conform to grade 304 of ASTM A 511.

* 3.2.4 Cast iron. Cast iron shall conform to the mechanical properties of ASTM A 48.

3.2.5 Brass.

3.2.5.1 Brass rod. Brass rod shall conform to 1/2 hard copper alloy No. 360 of QQ-B-626.

3.2.5.2 Brass wire. Brass wire shall conform to 1/2 hard of QQ-W-3ZI.

3.2.5.3 Brass castings. Brass casting shall conform to copper alloy No. 851-860 of QQ-C-390.

3.2.6 Brazing alloy. Brazing alloy shall conform to QQ-B-654.

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3.2.7 Gaskets.

3.2.7.1 Filler plug. The filler plug gasket shall be lead sheet conforming to grade B, of QQ-L-201,

3.2.7.2 Vent plug. The vent plug gasket shall be synthetic rubber conforming to SB615 AE5 of MIL-R-3065 and MIL-STD-417.

* 3.2.8 Nipple. The fuel supply line nipple shall conform to type I of WW-P-404.

3.2.9 Elbow. The fuel supply line elbow shall conform to type I of WW-P-521.

3.2.10 Safety chain. The brass safety chain shall conform to type II, class 6, 27 links per foot of RR-C-271.

3.2.11 Cotter pin. The cotter pin for the igniter assembly hinge shall be cadmium plated steel.

* 3.2.12 Stove pipe. Stove pipe shall have black rust inhibiting paint and shall conform to style 4, size 4-inch diameter of MIL-P-551. Four lengths of stove pipe shall be furnished with each immersion heater.

3.2.13 Primer. Primer shall conform to TT-P-636.

3.2.14 Enamel. Enamel shall conform to class A or B of TT-E-529. The color shall approximate Olive Drab 24087 of FED-STD-595.

3.2.15 Photographic plate, The photographic nameplate on the heater body and the instruction plate on the hood shall conform to type I, class 1 of GG-P-455 (see 6.5). The photographic nameplate and the caution plate on the fuel tank may be either plain or backed with a pressure-sensitive adhesive (see 6.6). The caution plate color shall be predominantly yellow.

3.3 Design and construction. The immersion heater components and assembly shall be constructed in accordance with Drawings 5-13-2679 through 5-13-2687 and requirements specified herein. Spot welds shall be adherent and well made so that the contact faces of the joint cannot be pulled apart without tearing the metal when tested as specified in 4.5.1. The nameplate on the fuel tank as shown on Drawing 5-13-2684 shall be applied smoothly and shall completely adhere to the tank.

3.3.1 Fuel tank. The fuel tank assembly shall be welded as specified except for the fuel line and vent valve bushings. The fuel line and vent valve bushings shall be brazed with brazing alloy specified in 3.2.6. The inside of the fuel tank assembly shall be cleaned of all foreign matter such as welding splatter, grease, and millings. The tank assembly prior to application of' finish shall not leak under 4 pounds + 1/2 pound per square inch, gage (psig) air pressure when tested in accordance with 4.5.2.

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3.3.2 Heater body. The heater body prior to application of finish shall not leak under 4 + 1/2 psig air pressure when tested in accordance with 4.5.3.

3.3.3 Igniter cup. Igniter cup shall not leak when tested in accordance with 4.5.4,

3.3.4 Filler plug. The filler plug gasket as specified in 3.2.7.1 shall be assembled to the filler plug with a press fit.

3.3.5 Valve assembly. The valve shall meter and stop the flow of fuel when tested in accordance with 4.5.6. The valve stern shall not bind throughout its entire engagement. The valve shall seat and stop the flow of air when closed with a torque of less than 4.0 inch pounds.

3.4 Welder qualification. The welders performing the welding on subassemblies, assemblies, and on the end item shall be certified welders and have passed the qualification test as prescribed by either the "Standard Qualification Procedures of the American Welding Society" or "Welding Qualification of the American Society of Mechanical Engineers" code. The contractor shall furnish the Government with a list of names of his welders who are certified and have passed the test of either referenced codes, and he shall certify that the welding of this equipment was performed by the welders listed .

3.5 Finish.

* 3.5.1 Zinc plating. Prior to assembly, zinc plating shall conform to type II, class Fe/Zn 13 Sc 3 of ASTM B 633.

3.5.2 priming and painting. prior to priming, surfaces of the heater assembly shall be cleaned in accordance with TT-C-490. Cleaned surfaces shall be coated with primer specified in 3.2.13, thoroughly dried and coated with enamel specified in 3.2.14. When class A enamel is used, the finish may be forced dried. When class B enamel is used, the enamel shall be baked. Touch-UP of bare spots and run-over inside of flue and burner tubes is acceptable. Surfaces of the heater assembly shall be painted as follows:

a. The heater body assembly down to a horizontal line 6 inches below the top of the burner tube that includes:

(1) The flue tube (prior to the assembly of the igniter assembly).
(2) The burner tube (prior to the assembly of the hood).
(3) The hanger support.
(4) The hanger assembly except that the hanger arm shall be completely painted exclusive of the screw threads.

b. All surfaces of the backing plate assembly.
c. Igniter assembly - All surfaces except wick and retainer spring.
d. Fuel tank - Except for the drip valve, all outside surfaces of the fuel tank including the filler plug and vent plug in the closed position.

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e. Hood - Prior to assembly of instruction plate, all surfaces of the hood including exposed surfaces of the hinges.

3.6 Instruction and nameplates. Instruction and nameplates shall consist of photographic plate specified in 3.2,15 (see 6.5). Dimensions and lettering shall conform to Drawing 5-13-2684.

3.7 Workmanship.

3.7.1 Fabrication. The immersion heater assembly shall be free of burrs, slivers, dents, and fractures. The fuel tank, when mounted on the heater, shall be in horizontal and vertical alignment with flue and burner tubes. The cover of the igniter assembly shall lay flush against the flue tube to completely cover the burner cup entrance hole.

3.7.2 Plating and painting. Plating and painting shall be smooth, adherent, free of blisters, cracks, runs, tackiness, embedded foreign material and areas of no finish.

3.7.3 Welding and brazing. The surface of the parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be smooth and free from cracks, burnholes, undercuts, splatter, or incomplete fusion. All scale and flux shall be removed from the welded area.

3.7.4 Spot welding. Spot welds, when applicable, shall be free of splatter and burn through and not otherwise poorly made.

3*7.5 Rivets. Rivets shall be tightly staked and shall not be broken.

3.8 Performance. The immersion heater, when operated in accordance with operating instructions on the nameplate (see Drawing 5-13-2684) and at an ambient temperature from 60° to 80°F, shall be capable of burning 86 to 94 antiknock index grade automotive gasoline (leaded or unleaded) to heat 24 gallons of water through a 100°F temperature rise within 35 minutes when tested as specified in 4.5.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

* 4,1.3 Certificates of compliance. When certificates of compliance are submitted , the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4,3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested as specified in 4.5.5. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

* 4.4.1.1 Certification. Certificate of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with specification requirements shall be examined and validated.

* 4.4.2 In-process inspection. Inspection shall be made of the following operations and requirements to establish conformance to specified requirements. Whenever nonconformance is noted, correction shall be made to the items affected and to the operation. Items which cannot be corrected shall be removed from production.

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4.4.2.1 Visual examination of heater surface preparation. Prior to enameling, examination shall be made of cleaning and priming of heater surfaces to establish conformance with requirements specified in 3.5.2. Whenever nonconformance is noted, correction shall be made to the process and all items processed.

4.4.2.2 Spot weld test. Prior to spot welding, the spot welding process shall be tested as specified in 4.5.1. Any failure to pass the test shall be cause for correction of the process.

4.4.2.3 Leak test. Prior to assembly, each fuel tank, heater body, and igniter cup shall be tested for leaks as specified in 4.5.2 through 4.5.4, respectively. Failure to pass any test shall be cause for rejection of the affected item.

4.4.3 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of immersion heaters. The sample unit shall be one heater. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE 1. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Missing, not as specified; finish coat not smooth, free of blisters, cracks, runs, tackiness, areas of no film and embedded foreign matter, or otherwise impaired		201
Materials, construction and workmanship (applicable to all components, subassemblies and assemblies)	Component material not as specified	101	
	Any part missing or not specified type	102	
	Fractured, split, punctured, torn, dented, creased, deteriorated, bowed, sprung, malformed, miscast	103	
	Component misplaced, leaking, or not in proper alignment	104	
	Evidence of sharp burrs, slivers, or edges	105	
	Any functioning component that requires abnormal force to operate		202

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Casting	Not connected or joined as specified or poorly accomplished	106	
Welding and brazing	Not smooth and free from cracks, burrholes, undercuts, splatter, or incomplete fusion	107	
	Not free from rust, flux, scale, paint, grease, or other foreign matter		203
spot welds	Not free of splatter and burn through, or otherwise poorly made	108	
Bolts, nuts, screws, studs , and other threaded fastener components	Missing, not type specified, broken, stripped, or fractured	109	
	Loose		204
Rivets	Missing, not tightly staked or broken	110	
Pipe, tubing, fittings and valves	Improperly connected, installed, or stripped	111	
Assembly	Any component not properly assembled or secured	112	
	Unit perceptibly out of square or alignment	113	
Identification and instruction marking	Missing, incomplete, illegible, not specified type or size, not in proper location, or not affixed in prescribed manner	114	
	Fuel tank nameplate not smooth or in complete adherence with tank		205

* 4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions annotated with asterisks on the drawings. Any annotated dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of immersion heaters. The sample unit shall be one heater. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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4.4.5 End item testing. When a first article is not required, the immersion heater shall be tested as specified in 4.5.5. Failure to pass the test shall be cause for rejection of the immersion heater.

* 4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Preservative application	Preservative improperly applied or missing
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container

4.5 Methods of inspection.

4.5.1 Spot weld test. Test coupons of the mating metals shall be joined together with spot welds in the same manner as intended for the parent components. The joint contact faces shall be' pulled apart to determine compliance with the spot weld requirements of 3.3. Any nonconformance shall constitute failure of this test.

4.5.2 Fuel tank test. The fuel tank shall be pressurized with air at 4 ± 1/2 psig and tested with a bubble supporting film in accordance with Method 5009 of FED-STD-101, to determine compliance with the leak requirements of 3.3.1. Any nonconformance shall constitute failure of this test.

4.5.3 Heater body test. The immersion heater body shall be pressurized with air at 4 + 1/2 psig and tested with a bubble supporting film in accordance with Method 5009 of FED-STD-101 to determine compliance with the leak requirements of 3.3.2. Any nonconformance shall constitute failure of this test. At the completion of this test the heater body shall be thoroughly dried.

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4.5.4 Igniter cup test. The igniter cup shall be filled with a fluid and observed for leaks to determine compliance with 3.3.3. Any nonconformance shall constitute failure of this test.

4.5.5 Performance test. The immersion heater shall be installed with the stack plumb in a 3Z gallon corrugated can conforming to RR-C-8Z and containing 24 gallons of water with a thermometer located in the center of the can opening and the bottom of the bulb immersed a minimum of 3 inches below the surface of the water. The heater shall be started with the fuel specified in 3.8 and operated at the steady fuel rate obtained by constantly adjusting the fuel valve so that smoke from the stack remains barely visible throughout this test. The valve shall meter and stop flow of fuel and not bind throughout its entire engagement as specified in 3.3.5. The time to heat the water shall be checked to determine compliance with the water heating requirements of 3.8. Any nonconformance shall constitute failure to pass this test.

4.5.6 Valve assembly test. The valve assembly will be pressurized at 4 + 1/2 psig to check for compliance with 3.3.5 to ensure that the valve shall seat and stop the flow of air when closed with a torque of less than 4 inch pounds as specified in 3.3.5. This test q ay be combined with 4.5.2.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A,

5.1.1.1 Disassembly. Each heater shall have the fuel tank detached from the body of the heater. The four stovepipe lengths shall be detached from the heater and disassembled. The burner shall be removed.

5.1.1.2 Cleaning. All uncoated ferrous metal surfaces of the heater and detached components shall be cleaned with dry-cleaning solvent conforming to P-D-680 and thoroughly dried prior to application of preservative.

5.1.1.3 Preservative application. The interior of the fuel tank and interior of the heater body shall be fogged with preservative conforming to type 1 of MIL-L-21260 and all excess preservative shall be drained off. All exterior uncoated surfaces of the heater body, including hanger and hanger screw threads, and fuel tank valve stem assembly shall be coated with preservative conforming to MIL-C-10382. The burner shall be completely wrapped with grease-proof barrier-material conforming to type II, grade A of MIL-B-121. The four cleaned lengths of the stovepipe shall be individually coated with q edium preservative oil conforming to MIL-L-3150.

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5.1.1.4 Unit packing.

5.1 .1.4.1 Heater body. The wrapped burner shall be placed within the burner flue and the hood shall be sealed in place over the burner flue with tape conforming to type IV, class 1 of PPP-T-60. The hanger screws shall be turned in, aligned, and secured to form the smallest package possible.

5.1.1.4.2 Fuel tank The valve assembly of the fuel tank shall be wrapped with grease-proof barrier-material specified in 5.1.1.3 and secured with tape as specified in 5.1.1.4.1.

5.1.1 .4.3 Stovepipe. The rivets used in the stovepipe lengths as retaining means shall be placed in a cotton cloth drawstring bag and securely tied to the outer stovepipe through one of the rivet holes. The preserved stovepipe lengths shall be nested one within the other and wrapped with grease-proof barrier-material specified in 5.1.1.3. The wrapped stovepipe lengths shall then be packed in a snug-fitting end-opening fiberboard box conforming to style FOL, W5C of PPP-B-636. The box shall be closed, waterproofed, and reinforced by means of tape in accordance with the appendix of PPP-B-636.

5.1 .1.4.4 Placement. The fuel tank shall be placed on end, the filler plug up, on top of the lower right portion of the heater body and against the flue stack so that the fuel tank support shall be located in one of the corners of the shipping container. The packed stovepipe shall be placed adjacent to the flue and the fuel tank on top of the lower left portion of the heater body. The tank and stovepipe length shall then be secured in place by not less than three bands of tape minimum 1/2 inch wide conforming to type IV of PPP-T-97. The tape shall be applied in a manner as to preclude loosening, movement, or damage to the secured components.

* 5.1.2 Commercial. Heaters shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Each heater, complete with components, preserved as specified in 5.1, shall be packed in a shipping container conforming to MIL-C-52950 and nailed wood container illustration shown on Drawing 5-13-1874, or a wirebound container conforming to PPP-B-585 and wirebound container illustration on Drawing 5-13-1874. Each heater shall be positioned within the shipping container with the flue in the same corner as the block provided on each container base. Heaters packed in wood crates shall be strapped to the base by means of a 3/4 by 0.023-inch flat steel strap conforming to class 1, type I of QQ-S-781. The container top shall be secured in place so that the block provided shall engage the exhaust flue tube opening. Each shipping container shall be closed and reinforced with flat steel strapping in accordance with the appendix of the applicable container document.

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* 5,2,2 Commercial packing. Heaters, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Marking. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6, NOTES

6.1 Intended use. The immersion heater is intended for heating or boiling water for cleaning kitchen utensils of field kitchens and mess kits and for other general uses.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When first article is required (see 3.1, 4.3, and 6.3).
co Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

6.2.1 Contract data requirements. Any data items to be delivered under any contract for items covered by this document should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

* 6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.2).

6.5 Photographic plates. Photographic plates for heater body and hood by Metalphoto Corporation, 18531 South Miles Road, Cleveland, OH 44128 or equal are satisfactory (see 3.2.15).

6.6 Pressure-sensitive plates. 3M Scotch Cal photosensitive plate number 8005 by the 3M Company, St. Paul, MN or equal is satisfactory for use as nameplate and caution plate with pressure sensitive adhesive, on the fuel tank. Industrial adhesives: Pliobond No. 20 by Goodyear Rubber and Tire Co., Akron, OH 44316; A851B by B. F. Goodrich General Products Company, Division of B. F. Goodrich Company, Akron, OH 43318 and Scotch-Grip by 3M Adhesives,

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Coatings and Sealers Division, 3M Center, St. Paul, MN 55101 or equal are satisfactory for use in adhering plain photographic nameplate on the fuel tank (see 3.2.15).

* 6.7 Subject term (key word) listing.

Field kitchens
Immersion heater
Liquid fuel heater
Water heating

6.8 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue..

Custodians:

Army - GL
Navy - YD

Review activities:

Army - MD
DLA - CS

User activity:

Navy - MC

Preparing activity:

Army - GL

Project No. 4540-0073

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-43540C	Heater, Immersion, M67, Liquid Fuel Fired, For Corrugated Cans

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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